WT901WIFI protocol

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1. Register table

ADDR (Hex)	ADDR (Dec)	REGISTER NAME	FUNCTION	SERIAL I/F	Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
00	00	SAVE	Save/Restart/Restore to Factory Defaults	R/W	SAVE[15:0]															
01	01	CALSW	Calibration Mode	R/W													CALSW[3:0]			
05	05	AXOFFSET	Acceleration X bias	R/W	AXOFFSET[15:0]															
06	06	AYOFFSET	Acceleration Y zero bias	R/W	AYOFFSET[15:0]															
07	07	AZOFFSET	Acceleration Z zero bias	R/W	AZOFFSET[15:0]															
08	08	GXOFFSET	Angular velocity X bias	R/W	GXOFFSET[15:0]															
09	09	GYOFFSET	Angular velocity Y bias	R/W	GYOFFSET[15:0]															
0A	10	GZOFFSET	Angular velocity Z zero bias	R/W	GZOFFSET[15:0]															
0B	11	HXOFFSET	Magnetic field X bias	R/W	HXOFFSET[15:0]															
0C	12	HYOFFSET	Magnetic field Y bias	R/W	HYOFFSET[15:0]															
0D	13	HZOFFSET	Magnetic field Z bias	R/W	HZOFFSET[15:0]															
1C	28	MAGRANGX	Magnetic field X calibration range	R/W	MAGRANGX[15:0]															
1D	29	MAGRANGY	Magnetic field Y calibration range	R/W	MAGRANGY[15:0]															
1E	30	MAGRANGZ	Magnetic field Z calibration range	R/W	MAGRANGZ[15:0]															
1F	31	BANDWIDTH	bandwidth	R/W													BANDWIDTH[3:0]			
20	32	GYRORANGE	Gyroscope range	R/W													GYRORANGE[3:0]			
twenty one	33	ACCRANGE	Acceleration range	R/W													ACCRANGE[3:0]			
twenty three	35	ORIENT	Installation direction	R/W																ORIENT
twenty four	36	AXIS6	algorithm	R/W																AXIS6

25	37	FILTK	Dynamic filtering	R/W	FILTK[15:0]
26	38	Q0	Quaternion 0	R	Q0[15:0]
27	39	Q1	Quaternion 1	R	Q1[15:0]
28	40	Q2	Quaternion 2	R	Q2[15:0]
29	41	Q3	Quaternion 3	R	Q3[15:0]
2A	42	ACCFILT	Acceleration filtering	R/W	ACCFILT[15:0]
2D	45	WIFIMODE	WIFI MODE	R	WIFIMODE[3:0]
2E	46	VERSION	Version Number	R	VERSION[15:0]
30	48	YYMM	years	R/W	MOUTH[15:8] YEAR[7:0]
31	49	DDH	Date and time	R/W	HOUR[15:8] DAY[7:0]
32	50	MMSS	Minutes and seconds	R/W	SECONDS[15:8] MINUTE[7:0]
33	51	MS	millisecond	R/W	MS[15:0]
34	52	AX	Acceleration X	R	AX[15:0]
35	53	AY	Acceleration Y	R	AY[15:0]
36	54	AZ	Acceleration Z	R	AZ[15:0]
37	55	GX	Angular velocity X	R	GX[15:0]
38	56	GY	Angular velocity Y	R	GY[15:0]
39	57	GZ	Angular velocity Z	R	GZ[15:0]
3A	58	HX	Magnetic FieldX	R	HX[15:0]
3B	59	HY	Magnetic field Y	R	HY[15:0]
3C	60	HZ	Magnetic field Z	R	HZ[15:0]
3D	61	Roll	Roll Angle	R	Roll[15:0]
3E	62	Pitch	Pitch Angle	R	Pitch[15:0]

3F	63	Yaw	Heading Angle	R	Yaw[15:0]
40	64	TEMP	temperature	R	TEMP[15:0]
42	66	BAT	Power	R	BAT[15:0]
43	67	TCPIP0	TCPIP	R	TCPIP0[15:0]
...					
4A	74	TCPIP7	TCPIP	R	TCPIP7[15:0]
4B	75	TCPPORT	TCP Port	R	TCPPORT[15:0]
4C	76	UDPIP0	UDPIP	R	UDPIP0[15:0]
...					
53	83	UDPIP7	UDPIP	R	UDPIP7[15:0]
54	84	UDPPORT	UDP Port	R	UDPPORT[15:0]
5D	93	RSSI	WIFI signal	R	RSSI[15:0]
60	96	LOCALPORT	Local Port	R	LOCALPORT[15:0]
68	104	RATE	Sending rate	R/W	RATE[15:0]
75	117	TCP/UDP	Connection	R/W	TCP/UDP
78	120	SSID0	WIFI ACCOUNT	R	SSID0[15:0]
...					
87	135	SSID15	WIFI ACCOUNT	R	SSID15[15:0]
88	136	PASSWORD0	WIFI PASSWORD	R	PASSWORD0[15:0]
...					
A7	167	PASSWORD31	WIFI PASSWORD	R	PASSWORD31[15:0]

2. Protocol Format

2.1 Reading Format

2.1.1 Data Analysis

- The data is sent in hexadecimal format, not ASCII.
- Each data is transmitted in sequence with low byte and high byte, and the two are combined into a signed short type data. For example, data DATA1, where DATA1L is the low byte and DATA1H is the high byte. The conversion method is as follows: Assuming that DATA1 is the actual data, DATA1H is its high byte part, and DATA1L is its low byte part, then: DATA1=(short)((short) DATA1H <<8| DATA1L). It must be noted here that DATA1H needs to be forced to be converted to a signed short type data before shifting, and the data type of DATA1 is also a signed short type, so that negative numbers can be represented.

Serial number	HEX code	ASCII Code	illustrate
1	0x57	W	Fixed, cannot be changed
2	0x54	T	Fixed, cannot be changed
3	0x35	5	Fixed, cannot be changed
4	0x35	5	Fixed, cannot be changed
5	ID		Set by the original manufacturer: ID: 0x30~0x39
6	ID		Set by the original manufacturer: ID: 0x30~0x39

7	ID		Set by the original manufacturer: ID: 0x30~0x39
8	ID		Set by the original manufacturer: ID: 0x30~0x39
9	ID		Set by the original manufacturer: ID: 0x30~0x39
10	ID		Set by the original manufacturer: ID: 0x30~0x39
11	ID		Set by the original manufacturer: ID: 0x30~0x39
12	ID		Set by the original manufacturer: ID: 0x30~0x39, device ID is 12, for example WT5500001234
13	YY		Year
14	MM		moon
15	DD		day
16	HH		hour
17	MM		point
18	SS		Second
19	MSL		Millisecond low
20	MSH		Millisecond high, millisecond = (MSH << 8) | MSL
twenty one	AXL		Acceleration X low
twenty	AXH		Acceleration X high, acceleration X =

two			((AXH<<8)|AXL)/32768*16 (g)
twenty three	AYL		Acceleration Y low
twenty four	YYH		Acceleration Y high, acceleration Y = ((AYH<<8)|AYL)/32768*16 (g)
25	AZL		Acceleration Z Low
26	AZH		Acceleration Z high position, acceleration Z = ((AZH<<8)|AZL)/32768*16 (g)
27	GXL		Angular velocity X low
28	HkDJ		Angular velocity X high position, angular velocity X = ((GXH<<8)|GXL)/32768*2000 (°/s)
29	G J		Angular velocity Y low
30	G JZ		Angular velocity Y high position, angular velocity Y = ((GYH<<8)|GYL)/32768*2000 (°/s)
31	GZL		Angular velocity Z low
32	G JZ		Angular velocity Z high position, angular velocity Z = ((GZH<<8)|GZL)/32768*2000 (°/s)
33	HXL		Magnetic field X low
34	HkDJ		Magnetic field X high position, magnetic field X = ((HXH<<8)|HXL)*100/1024 (uT)
35	HYL		Magnetic field Y low

36	HYH		Magnetic field Y high position, magnetic field Y = ((HYH<<8)|HYL)*100/1024 (uT)
37	HkDJ		Magnetic field Z low position
38	HkDJ		Magnetic field Z high position, magnetic field Z = ((HZH<<8)|HZL)*100/1024 (uT)
39	AgX		Angle X Low
40	HkDJ		Angle X high position, Angle X = ((AgXH<<8)|AgXL)/32768*180 (°)
41	oeLh		Angle Y low
42	YG		Angle Y high position, Angle Y = ((AgYH<<8)|AgYL)/32768*180 (°)
43	G JZ		Angle Z Low
44	G JZ		Angle Z high position, Angle Z = ((AgZH<<8)|AgZL)/32768*180 (°)
45	TEMPL		Low temperature
46	TEMPH		Temperature high, temperature = ((TEMPH <<8) | TEMPL) / 100 (℃)
47	BATL		Low battery
48	BATH		High power level, power level = ((BATH <<8)| BATL)/100 (V)
49	RSSIL		Signal low

50	RSSIH		Signal high, signal = (RSSIH <<8) | RSSIL (dBm)
51	VERSIONL		Version number low
52	VERSION H		High bit of the version number, version number = (VERSIONH <<8)| VERSIONL
53		0x0D	Line Breaks
54		0x0A	Line Breaks

2.1.2 Reading register return value

Used to read the value of the user-specified register. Reading REG1 returns the values of the four registers REG1 to REG4. The protocol must return four registers.

Send command: FF AA 27 REG 00

The return is as follows:

0x55	0x5F	REG1L	REG1H	REG2L	REG2H	REG3L	REG3H	REG4L	REG4H	SUM
name	describe	Remark								
REG1L	Register 1 lower 8 bits	REG1[15:0]=((REG1H<<8)|REG1L)								
REG1H	Register 1 high 8 bits									
REG2L	Register 2 lower 8 bits	REG2[15:0]=((REG2H<<8)|REG2L)								
REG2H	Register 2 high 8 bits									
REG3L	Register 3 lower 8 bits	REG3[15:0]=((REG3H<<8)|REG3L)								
REG3H	Register 3 high 8 bits									
REG4L	Register 4 lower 8 bits	REG4[15:0]=((REG4H<<8)|REG4L)								
REG4H	Register 4 high 8 bits									

SUM	Checksum	SUM=0x55+0x5F+REG1L+REG1H+REG2L+REG2H+REG3L+REG3H+REG4L+REG4H

For example:

Reading register "AXOFFSET" returns: 0x55 0x5F AXOFFSET[7:0] AXOFFSET[15:8] AYOFFSET[7:0] AYOFFSET[15:8] AZOFFSET[7:0] AZOFFSET[15:8] GXOFFSET[7:0] GXOFFSET[15:8] SUM

2.2 Writing Format

2.2.1 ASCII setting format

Function	instruction	illustrate
Calibration time	WTTIME:20230728125959001\r\n	Calibrate sensor time Year: 2023 Month: 07 Day: 28 Hours: 12 Points: 59 Sec: 59 Milliseconds: 001
UDP Broadcast IP	WIT192.168.4.2\r\n	192.168.4.2: Determined by the user's computer Go to IP: 255.255.255.255, port 9250 to broadcast the IP. After the sensor searches for the IP, it will immediately switch the IP.
Setting up Wi-Fi	WITWIFI:"WIT_A","wtznwtzn"\r\n	WIT_A: User WIFI name wtznwtzn: User WIFI password WIFI must be in the 2.4GHz frequency band. After setting, it will automatically switch to STA mode (WIFIMODE register)
Designated user TCP (STA mode TCP)	TCPIP:192.168.4.2,1399\r\n	192.168.4.2: Determined by the user 1399: According to user decision This function is only supported in STA mode, so after setting, it will automatically switch to STA mode TCP communication (WIFIMODE register), and the data will be sent to the server specified by the user. In STA mode, the user's router WIFI (2.4GHz band) must be connected

Designated user UDP (STA mode UDP)	UDPIP:192.168.4.2,1399\r\n	192.168.4.2: Determined by the user 1399: According to user decision This function is only supported in STA mode, so after setting, it will automatically switch to STA mode UDP communication (WIFIMODE register), and the data will be sent to the server specified by the user. In STA mode, the user's router WIFI (2.4GHz band) must be connected
Witt TCP	WITTCP\r\n	Connect to Vetter Smart's cloud platform
AP Mode TCP	WITAPTCP\r\n	After setting, it will automatically switch to TCP communication in AP mode. The sensor will generate WIFI, and the computer can connect to the WIFI. The WIFI name is the device ID.
AP Mode UDP	WITAPUDP\r\n	After setting, it will automatically switch to UDP communication in AP mode. The sensor will generate WIFI, and the computer can connect to the WIFI. The WIFI name is the device ID.
Set up TCPIP and WIFI at the same time	IPWIFI:"WIT_A","wtznwtzn";TCP192.168.4.2,1399\r\n	Set IP and WIFI at the same time. If the user changes IP or WIFI remotely, it is recommended to use this command to avoid sending WIFI and IP settings separately. Otherwise, the IP change will take effect and the data will be sent to other places, and the remote modification will not be possible. Therefore, it is recommended to use this command
Set up UDPIP and WIFI at the same time	IPWIFI:"WIT_A","wtznwtzn";UDP192.168.4.2,1399\r\n	Set IP and WIFI at the same time. If the user changes IP or WIFI remotely, it is recommended to use this command to avoid sending WIFI and IP settings separately. Otherwise, the IP change will take effect and the data will be sent to other places, and the remote modification will not be possible. Therefore, it is recommended to use this command

2.2.2 Write register format (HEX code format)

- This method is more flexible, and each register can be operated, but it is easy to set it wrong. You need to understand each register before setting it, and it only supports single setting. For multiple settings (account password), you need to send multiple registers, which wastes a lot of time for data exchange.
- The following data are all in Hex code
- All settings require unlocking the register (KEY) first.

Protocol Header	Protocol Header	register	Data lower 8 bits	Data high 8 bits
0xFF	0xAA	ADDR	DATAL[7:0]	DATAH[15:8]

- The data is sent in hexadecimal format, not ASCII.
- Each data is transmitted in sequence as low byte and high byte, and the two are combined into a signed short type data. For example, data DATA, where DATAL is the low byte and DATAH is the high byte. The conversion method is as follows: Assuming DATA is the actual data, DATAH is its high byte part, and DATAL is its low byte part, then: DATA=(short)((short) DATAH <<8| DATAL). It must be noted here that DATAH needs to be forced to be converted to a signed short type data before shifting, and the data type of DATA is also a signed short type, so that negative numbers can be represented.

For example:

Acceleration calibration (the module must be placed on the front for acceleration calibration. If the module is placed on the back for calibration, it will cause abnormal acceleration, which will lead to abnormal angle)

1. Enter calibration: FF AA 01 01 00 (calibration uses register 0x01, which will be discussed later)

2. Delay 5 seconds

3. Exit calibration: FF AA 01 00 00

4. Save calibration: FF AA 00 00 00

2.2.2.1 SAVE (Save/Restart/Restore to Factory)

Register Name: SAVE

Register address: 0 (0x00)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	SAVE[15:0]	Save: 0x0000 Restart: 0x00FF Factory Reset: 0x0001

Example: FF AA 00 FF 00 (restart)

Example: FF AA 00 00 00 (save)

Example: FF AA 00 01 00 (factory restore)

2.2.2.2 CALSW (calibration mode)

Register Name: CALSW

Register address: 1 (0x01)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	CAL[3:0]	To set the calibration mode: 0000(0x00): Normal working mode 0001(0x01): Automatic addition calibration 0011(0x03): Clear the height (this version does not have this function) 0100(0x04): Heading angle reset to zero 0111(0x07): Magnetic field calibration (spherical fitting method) 1000(0x08): Set angle reference 1001(0x09): Magnetic field calibration (dual plane mode, this version does not have this function)

Example: FF AA 01 04 00 (heading angle set to zero)

Add-on calibration example:

FF AA 01 01 00 (Enter the calibration, please make sure it is placed horizontally)

Wait for 5 seconds, the automatic calibration is completed

FF AA 01 00 00 (Exit calibration)

FF AA 00 00 00 (save)

Magnetic field calibration example:

FF AA 01 07 00 (enter magnetic field calibration)

Rotate 1~2 times around the north on each axis. Please refer to the magnetic field calibration video

FF AA 01 00 00 (Exit magnetic field calibration)

FF AA 00 00 00 (save)

2.2.2.3 AXOFFSET~HZOFFSET (zero offset setting)

Register name: AXOFFSET~ HZOFFSET

Register address: 5~13 (0x05~0x0D)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AXOFFSET[15:0]	Acceleration X-axis zero bias, actual acceleration zero bias = AXOFFSET[15:0]/10 (LSB)
15:0	AYOFFSET[15:0]	Acceleration Y-axis zero offset, actual acceleration zero offset = AYOFFSET[15:0]/10(LSB)
15:0	AZOFFSET[15:0]	Acceleration Z-axis zero offset, actual acceleration zero offset = AZOFFSET[15:0]/10 (LSB)
15:0	GXOFFSET[15:0]	Angular velocity X-axis zero bias, actual angular velocity zero bias = GXOFFSET[15:0]/10(LSB)

15:0	GYOFFSET[15:0]	Angular velocity Y-axis zero bias, actual angular velocity zero bias = GYOFFSET[15:0]/10(LSB)
15:0	GZOFFSET[15:0]	Angular velocity Z-axis zero bias, actual angular velocity zero bias = GZOFFSET[15:0]/10(LSB)
15:0	HXOFFSET[15:0]	Magnetic field X-axis zero bias
15:0	HYOFFSET[15:0]	Magnetic field Y-axis zero bias
15:0	HZOFFSET[15:0]	Magnetic field Z-axis zero bias
Example: FF AA 05 E8 03 (set acceleration X-axis zero bias to 0.0488g), 0x03E8=1000, 1000/10=100 (LSB), the user can automatically add calibration; if you want to set it yourself, read the original acceleration, multiply it by 10 times and then send it.		

2.2.2.4 MAGRANGX~MAGRANGZ (magnetic field calibration range)

Register name: MAGRANGX~MAGRANGZ

Register address: 28~30 (0x1C~0x1E)

Read/write direction: R/W

Default value: 0x01F4

Bit	NAME	FUNCTION
15:0	MAGRANGX[15:0]	Magnetic field calibration X-axis range
15:0	MAGRANGY[15:0]	Magnetic field calibration Y-axis range
15:0	MAGRANGZ[15:0]	Magnetic field calibration Z-axis range
Example: FF AA 1C F4 01 (set the magnetic field calibration X-axis range to 500)		

2.2.2.5 BANDWIDTH

Register Name: BANDWIDTH

Register address: 31 (0x1F)

Read/write direction: R/W

Default value: 0x0004

Bit	NAME	FUNCTION
15:4		
3:0	BANDWIDTH[3:0]	Setting bandwidth 0000(0x00) : 256Hz 0001(0x01) : 188Hz 0010(0x02) : 98Hz 0011(0x03) : 42Hz 0100(0x04) : 20Hz 0101(0x05) : 10Hz 0110(0x06) : 5Hz

Example: FF AA 1F 01 00 (set bandwidth to 188 Hz)

2.2.2.6 GYRORANGE (Gyroscope range)

Register Name: GYRORANGE

Register address: 32 (0x20)

Read/write direction: R/W

Default value: 0x0003

Bit	NAME	FUNCTION
15:4		

3:0	GYRORANGE[3:0]	Setting the gyroscope range 0011(0x00) : 250°/s 0011(0x01) : 500°/s 0011(0x02) : 1000°/s 0011(0x03) : 2000°/s (default)
Example: FF AA 20 03 00 (set the gyro range to 2000°/s)		

2.2.2.7 ACCRANGE (accelerometer range)

Register Name: ACCRANGE

Register address: 33 (0x21)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:4		
3:0	ACCRANGE[3:0]	Setting the accelerometer range 0000(0x00) : ±2g 0000(0x01) : ±4g 0000(0x02) : ±8g 0011(0x03) : ±16g (default)
Example: FF AA 21 00 00 (set the accelerometer range to 2g)		

2.2.2.8 ORIENT (Installation direction)

Register Name: ORIENT

Register address: 35 (0x23)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:1		
0	ORIENT	Set the installation direction 0(0x00) : Horizontal installation 1(0x01) : Vertical installation (the Y-axis arrow of the coordinate axis must face upward)

Example: FF AA 23 01 00 (set for vertical installation)

2.2.2.9 AXIS6 (algorithm)

Register name: AXIS6

Register address: 36 (0x24)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:1		
0	AXIS6	Setting up the algorithm 0(0x00): 9-axis algorithm (magnetic field solution navigation angle, absolute heading angle) 1(0x01): 6-axis algorithm (integral solution of navigation angle, relative heading angle)

Example: FF AA 24 01 00 (set 6-axis algorithm mode)

2.2.2.10 FILTK (K value filtering)

Register Name: FILTK

Register address: 37 (0x25)

Read/write direction: R/W

Default value: 0x001E

Bit	NAME	FUNCTION
15:0	FILTK[15:0]	Range: 1~10000, default 30 (modification is not recommended. If the angle does not meet the use requirements after modification, please change it to 30) The smaller FILTK[15:0] is, the more reliable the angular velocity data is, the stronger the anti-vibration performance is, and the weaker the real-time performance is. If all angular velocities are taken, set it to 1. The larger the FILTK[15:0] is, the more reliable the acceleration data is, the weaker the anti-vibration performance is, and the stronger the real-time performance is. If all accelerations are taken, set it to 10000.

Example: FF AA 25 1E 00 (set the K value filter to 30)

2.2.2.11 Q0~Q3 (quaternion)

Register name: Q0~Q3

Register address: 38~41 (0x26~0x29)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	Q0[15:0]	Quaternion 0 = Q0[15:0]/32768
15:0	Q1[15:0]	Quaternion 1 = Q1[15:0]/32768
15:0	Q2[15:0]	Quaternion 2 = Q2[15:0]/32768
15:0	Q3[15:0]	Quaternion 3 = Q3[15:0]/32768

2.2.2.12 ACCFILT (acceleration filter)

Register Name: ACCFILT

Register address: 42 (0x2A)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	ACCFILT[15:0]	Range: 1~10000, default 500 (modification is not recommended. If the angle does not meet the use requirements after modification, please change it to 0) The smaller ACCFILT[15:0] is, the stronger the anti-vibration performance is and the weaker the real-time performance is. The larger the ACCFILT[15:0], the weaker the anti-seismic performance and the stronger the real-time performance. This parameter is an empirical value and needs to be adjusted according to different environments. In the tractor environment, ACCFILT[15:0] can be adjusted to 300, because the tractor shakes severely and needs to improve the anti-vibration performance

Example: FF AA 2A F4 01 (set acceleration filter to 500)

2.2.2.13 WIFIMODE

Register name: WIFIMODE

Register address: 45 (0x2D)

Read/write direction: R

Default value: 0x0002

Bit	NAME	FUNCTION

15:4		
3:0	WIFIMODE	Others: STA mode (need to connect to 2.4G WIFI) 0010(0x02): AP mode (sensor broadcasts WIFI, equivalent to routing)

2.2.2.14 VERSION

Register Name: VERSION

Register address: 46 (0x2E)

Read/write direction: R

Default value: 0x32D3(13011)

Bit	NAME	FUNCTION
15:0	VERSION[15:0]	13011

Example:

Send: FF AA 27 2E 00 (read version number, 0x27 means read, 0x2E is the version number register)

Return: 55 5F VL VH XX XX XX XX XX XX SUM

VERSION[15:0]=(short)(((short)VH<<8)|VL)

2.2.2.15 YYMM~MS (on-chip time)

Register name: YYMM~MS

Register address: 48~51 (0x30~0x33)

Read/write direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:8	YYMM[15:8]	moon

7:0	YYMM[7:0]	Year
15:8	DDHH[15:8]	hour
7:0	DDHH[7:0]	day
15:8	MMSS[15:8]	Second
7:0	MMSS[7:0]	point
15:0	MS[15:0]	millisecond

Example:

FF AA 30 16 03 (set year and month 22-03)

FF AA 31 0C 09 (set date and time 12-09)

FF AA 32 1E 3A (set minutes and seconds 30:58)

FF AA 33 F4 01 (set milliseconds 500)

Example:

Send: FF AA 27 30 00 (read version number, 0x27 means read, 0x30 is year and month register)

Return: 55 5F YYMM[7:0] YYMM[15:8] DDHH[7:0] DDHH[15:8] MMSS[7:0] MMSS[15:8] MS[7:0] MS[15:8] SUM

2.2.2.16 AX~AZ (acceleration)

Register name: AX~AZ

Register address: 52~54 (0x34~0x36)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AX[15:0]	Acceleration X = AX[15:0]/32768*16g (g is the acceleration due to gravity, which can be 9.8m/s2)

15:0	AY[15:0]	Acceleration Y = AY[15:0]/32768*16g (g is the acceleration due to gravity, which can be 9.8m/s2)
15:0	AZ[15:0]	Acceleration Z = AZ[15:0]/32768*16g (g is the acceleration due to gravity, which can be 9.8m/s2)
The 16 in the formula is determined by the range set by the user, the default is 16g		

2.2.2.17 GX~GZ (angular velocity)

Register name: GX~GZ

Register address: 55~57 (0x37~0x39)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	GX[15:0]	Angular velocity X = GX[15:0]/32768*2000°/s
15:0	GY[15:0]	Angular velocity Y = GY[15:0]/32768*2000°/s
15:0	GZ[15:0]	Angular velocity Z = GZ[15:0]/32768*2000°/s
The 2000 in the formula is determined by the range set by the user, and the default is 2000°/s		

2.2.2.18 HX~HZ (magnetic field)

Register name: HX~HZ

Register address: 58~60 (0x3A~0x3C)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	HX[15:0]	Magnetic field X = HX[15:0]*100/1024 (unit: uT)

15:0	HY[15:0]	Magnetic field Y = HY[15:0] * 100/1024 (unit: uT)
15:0	HZ[15:0]	Magnetic field Z = HZ[15:0] *100/1024 (unit: uT)

2.2.2.19 Roll~Yaw (angle)

Register Name: Roll~Yaw

Register address: 61~63 (0x3D~0x3F)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	Roll[15:0]	Roll angle X = Roll[15:0]/32768*180°
15:0	Pitch[15:0]	Pitch angle Y = Pitch[15:0]/32768*180°
15:0	Yaw[15:0]	Heading angle Z = Yaw[15:0]/32768*180°

2.2.2.20 TEMP (temperature)

Register Name: TEMP

Register address: 64 (0x40)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	TEMP[15:0]	Temperature = TEMP[15:0]/100°C

2.2.2.21 BAT (battery capacity)

Note: The battery is out of power when the power level is <= 3.3.

Register name: BAT

Register address: 66 (0x42)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	BAT[15:0]	Battery power = BAT[15:0]/100(V)

the corresponding relationship between voltage and percentage of electricity is:

register Value	voltage value	percentage of electricity
>396	>3.96V	100 percent
393-396	3.93V-3.96V	90%
387-393	3.87V-3.93V	75%
382-387	3.82V-3.87V	60%
379-382	3.79V-3.82V	50%
377-379	3.77V-3.79V	40%
373-377	3.73V-3.77V	30%
370-373	3.70V-3.73V	20%
368-370	3.68V-3.70V	15%
350-368	3.50V-3.68V	10%
340-350	3.40V-3.50V	5%
<3.40V	<3.40V	0%

2.2.2.22 TCPIP0~TCPIP7 (TCPIP)

Register name: TCPIP0~TCPIP7

Register address: 67~74 (0x43~0x4A)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
63:0	TCPIP[63:0]	Default: 192.168.4.2

2.2.2.23 TCPPORT (TCP remote port)

Register Name: TCPPORT

Register address: 75 (0x4B)

Read/write direction: R

Default value: 0x0577

Bit	NAME	FUNCTION
15:0	TCPPORT[15:0]	Default port: 1399

2.2.2.24 UDPIP0~UDPIP7 (UDPIP)

Register name: UDPIP0~UDPIP7

Register address: 76~83 (0x4C~0x53)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
63:0	UDPIP[63:0]	Default: 192.168.4.2

2.2.2.25 UDPPORT (UDP remote port)

Register Name: UDPPORT

Register address: 84 (0x54)

Read/write direction: R

Default value: 0x0577

Bit	NAME	FUNCTION
15:0	UDPPORT[15:0]	Default port: 1399

2.2.2.26 RSSI (WIFI signal)

Register Name: RSSI

Register address: 93 (0x5D)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	BAT[15:0]	Temperature = BAT[15:0]/10(V)

2.2.2.27 LOCALPORT

Register Name: LOCALPORT

Register address: 96 (0x60)

Read/write direction: R

Default value: 0x2422

Bit	NAME	FUNCTION
15:0	LOCALPORT[15:0]	Default port: 9250

2.2.2.28 RATE (transmission rate)

Register Name: RATE

Register address: 104 (0x68)

Read/write direction: R/W

Default value: 0x0064

Bit	NAME	FUNCTION

15:0	RATE[15:0]	Default: 100ms, 1000/100ms = 10Hz
Example (setting 200Hz): FF AA 68 05 00，1000/5ms=200Hz		

2.2.2.29 TCP/UDP (connection mode)

Register Name: TCP/UDP

Register address: 104 (0x68)

Read/write direction: R/W

Default value: 0x0064

Bit	NAME	FUNCTION
15:1		
0	TCP/UDP	0: UDP (default) 1: TCP

2.2.2.30 SSID0~SSID15 (WIFI account)

Register name: SSID0~SSID15

Register address: 120~135 (0x78~0x87)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
127:0	SSID[128:0]	

2.2.2.31 PASSWORD0~PASSWORD31 (WIFI password)

Register name: PASSWORD0~PASSWORD31

Register address: 136~167 (0x88~0xA7)

Read/write direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
255:0	PASSWORD[255:0]	